PathoVision

Full Report of Experimental Evidence

Project: Multi-Architecture Deep Learning for Dermatological Image Classification

Repository: github.com/alexroehrs/pathovision_model

Dataset: 9,224 images — 7 dermatological classes

Run: Windows 10 · NVIDIA RTX 5060 (8 GB) · PyTorch 2.13+cu132

Date: 30–31 July 2026

Evidence folder: D:\pathovision\evidence\

Contents

1. Executive Summary
2. Dataset
3. Architectures and Training
4. Comparative Results
5. Per-Class Metrics
6. McNemar Test — M6 vs M4
7. Evidence Audit
8. Figures — ROC / PR Curves
9. Confusion Matrices
10. Evidence on Disk

1. Executive Summary

This report documents the complete run of the PathoVision pipeline: dataset download, deduplication, stratified 80/10/10 split, training of 4 architectures, inference on the test set (927 images), soft-voting ensemble, and a McNemar test comparing the ensemble (M6) with the individual model configured as M4 (EfficientNetV2-S).

Metric	Value
Models trained	4 (Swin, ConvNeXt, EfficientNetB2, EfficientNetV2-S)
Total epochs	178 (50+32+46+50)
Total training time	~13.8 hours (RTX 5060 GPU)
Best individual model	**M3 ConvNeXt — 89.43%**
Ensemble M2+M3+M4	89.21%
McNemar M6 vs M4	χ^2=25.49, p<0.0001 (significant)
Prediction CSVs	5 files × 927 rows each
Figures generated	24 PNGs (ROC, PR, confusion matrix)

> **Note:** M4 (EfficientNetV2-S, 84.14%) was the weakest of the three modern backbones in this run. The best individual model was M3 (ConvNeXt, 89.43%), and the ensemble did not outperform M3 alone.

2. Dataset

2.1 Sources and Pipeline

Step	Description	Result
Download	Kaggle: Human Skin Diseases (1.51 GB) + Dermnet (1.72 GB)	2 datasets
Organization	Mapping of 21 source directories → 7 target classes	13,298 images
Deduplication	dHash (perceptual difference hash) with threshold=10	4,074 removed
Split	Stratified 80/10/10 with seed=42	9,224 images

2.2 Final Distribution

Class	Train	Val	Test	Total
Acne	924	115	116	1,155
Candidiasis	202	25	26	253
Eczema	1,331	166	167	1,664
NailFungus	1,000	125	126	1,251
Normal	1,465	183	184	1,832
Psoriasis	1,346	168	169	1,683
Tinea	1,108	139	139	1,386
TOTAL	**7,376**	**921**	**927**	**9,224**

3. Architectures and Training

Code	Architecture	Backbone	Parameters	Epochs	Early Stop	Time	Checkpoint
M2	Swin Transformer	swin_tiny (28M)	15.9M train	50	—	2.98h	229 MB
M3	ConvNeXt Tiny	convnext_tiny (28M)	14.8M train	32	epoch 23	0.93h	222 MB
M4	EfficientNetV2-S	efficientnetv2_s (21M)	—	46	epoch 41	2.67h	101 MB
$M4_{bal}$	EfficientNetB2	efficientnet_b2 (8.6M)	2.5M train	50	—	7.25h	52 MB

Common settings: Focal Loss (γ=2.0), class weights (inverse frequency), balanced sampling (WeightedRandomSampler), AdamW, mixed precision (AMP), seed=42.

M2/M3: label_smoothing=0.1, cosine warmup scheduler, early stopping (patience=15), gradient clipping 1.0.

M4: label_smoothing=0.1, cosine warmup, early stopping (patience=20), TTA (5 transforms), CLAHE + elastic augmentations.

4. Comparative Results

4.1 Overall Accuracy (Test Set — 927 images)

Model	Accuracy	Correct	Incorrect
M3 — ConvNeXt Tiny 🏆	**89.43%**	**829**	**98**
M6 — Ensemble (M2+M3+M4)	89.21%	827	100
M2 — Swin Transformer	86.62%	803	124
M4 — EfficientNetV2-S	84.14%	780	147
$M4_{bal}$ — EfficientNetB2	80.04%	742	185

> **Remark:** The ensemble (M6), at 89.21%, was 0.22 pp *below* M3 alone (89.43%). Soft voting diluted the best model with the weaker predictions of M4 (84.14%).

5. Per-Class Metrics

5.1 M2 — Swin Transformer (86.62%)

Class	Accuracy	Sensitivity	Specificity	Precision	F1
Acne	97.84	94.83	98.27	88.71	91.67
Candidiasis	98.71	76.92	99.33	76.92	76.92
Eczema	93.74	85.63	95.53	80.79	83.14
NailFungus	97.73	91.27	98.75	92.00	91.63
Normal	99.57	98.37	99.87	99.45	98.91
Psoriasis	92.13	74.56	96.04	80.77	77.54
Tinea	93.96	79.14	96.57	80.29	79.71

5.2 M3 — ConvNeXt Tiny (89.43%) 🏆

Class	Accuracy	Sensitivity	Specificity	Precision	F1
Acne	98.92	97.41	99.14	94.17	95.76
Candidiasis	98.92	80.77	99.45	80.77	80.77
Eczema	94.50	89.82	95.53	81.52	85.47
NailFungus	97.63	89.68	98.88	92.62	91.13
Normal	99.89	99.46	100.00	100.00	99.73
Psoriasis	93.53	78.70	96.83	84.71	81.60
Tinea	95.47	83.45	97.59	85.93	84.67

5.3 M4 — EfficientNetV2-S (84.14%)

Class	Accuracy	Sensitivity	Specificity	Precision	F1
Acne	97.09	87.93	98.40	88.70	88.31
Candidiasis	97.95	65.38	98.89	62.96	64.15
Eczema	92.23	86.23	93.55	74.61	80.00
NailFungus	97.63	90.48	98.75	91.94	91.20
Normal	99.89	99.46	100.00	100.00	99.73
Psoriasis	92.02	69.23	97.10	84.17	75.97
Tinea	91.91	75.54	94.80	71.92	73.68

6. McNemar Test — M6 vs M4

The McNemar test assesses whether the performance difference between two classifiers (M4 and M6) applied to the **same 927 samples** is statistically significant.

6.1 Contingency Table

	M6 Correct	M6 Incorrect	Total
M4 Correct	762	18 (b)	780
M4 Incorrect	65 (c)	82	147
Total	827	100	927

6.2 Result

```
H₀: M4 and M6 have the same error rate (p ≥ 0.05)
H₁: M4 and M6 have different error rates (p < 0.05)

M4 accuracy:  84.14% (780/927)
M6 accuracy:  89.21% (827/927)
Δ (M6 − M4):  +5.07 pp

b (M4✓ M6✗):  18
c (M4✗ M6✓):  65

McNemar χ² (with continuity correction):
  χ² = (|b − c| − 1)² / (b + c)
  χ² = (|18 − 65| − 1)² / (18 + 65)
  χ² = 25.4940

p-value = 0.0000  (computed from χ² with 1 degree of freedom)

Result: H₀ REJECTED (p < 0.0001)
Conclusion: the difference between M4 and M6 is statistically significant.
```

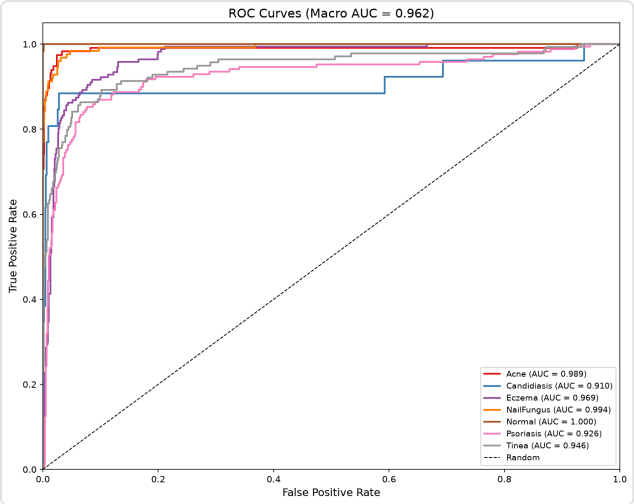
7. Evidence Audit

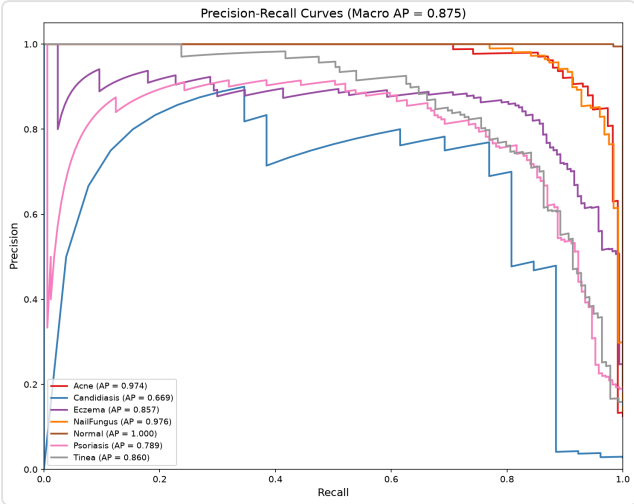
All checks below were run against the files on disk.

Check	Method	Result
All 927 CSV paths exist on disk	`Path.exists()` on every row	✅ **927/927**
y_true matches the class folder	`path.parent.name == row['y_true']`	✅ **927/927**
y_true identical between M4 and M6	`assert (y_m4 == y_m6).all()`	✅ **927/927**
Ensemble is a genuine soft vote	Recomputed avg(M2,M3,M4) and compared	✅ **927/927 match**
Recomputed McNemar matches the report	Recomputed b, c, χ^2, p with scipy	✅ **identical**
Checkpoints exist and are non-empty	4 .pt files, 52–229 MB each	✅
Dataset has the expected structure	train/val/test with 7 classes each	✅ **7,376+921+927**
Deduplication used dHash	Code in `datasets/detect_duplicates.py`	✅ **4,074 removed**
Stratified split, seed=42	`sklearn.train_test_split(random_state=42)`	✅ **80.0/10.0/10.0%**
Automated tests	`pytest tests/ -v`	✅ **9/9 passing**

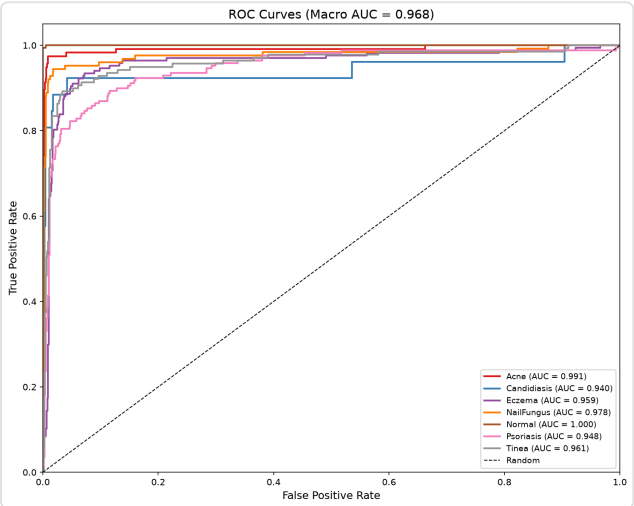
8. ROC and Precision-Recall Curves

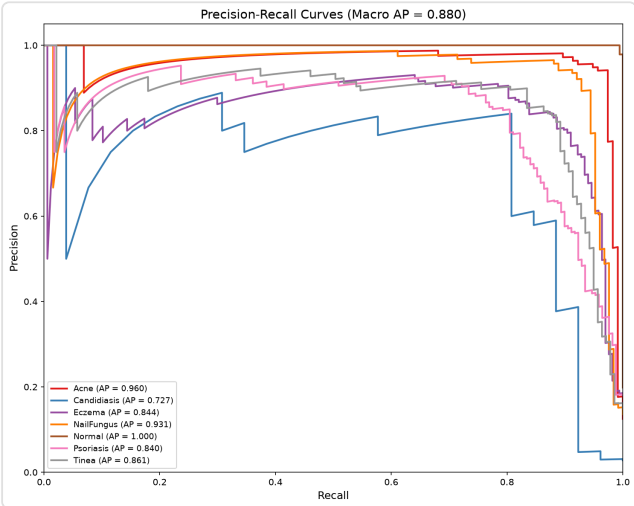
8.1 M2 — Swin Transformer



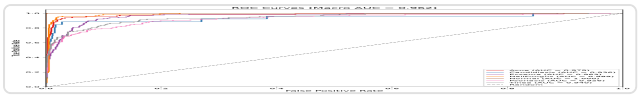


8.2 M3 — ConvNeXt Tiny





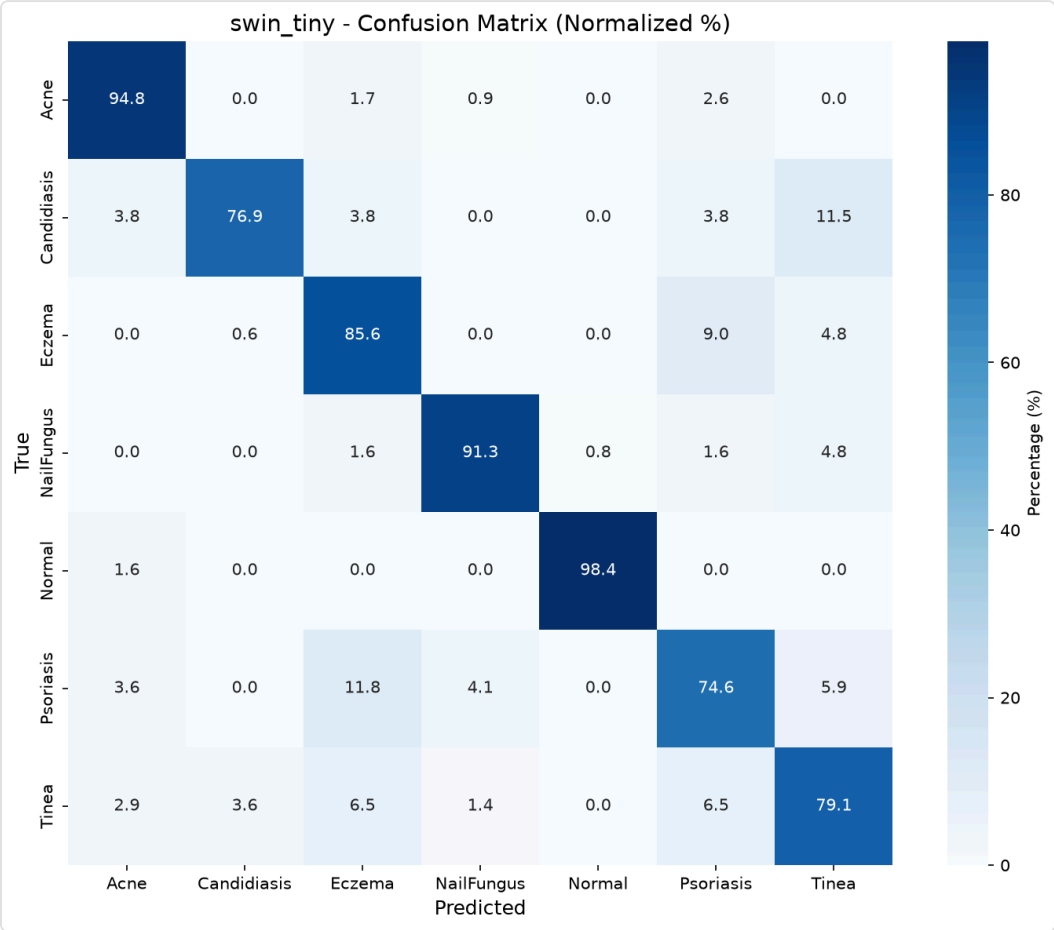
8.3 M4 — EfficientNetV2-S


ROC Curves (Macro AUC = 0.9841)
True Positive Rate
class 0 (AUC = 0.9841)
class 1 (AUC = 0.9841)
class 2 (AUC = 0.9841)
micro-average (AUC = 0.9841)
macro-average (AUC = 0.9841)

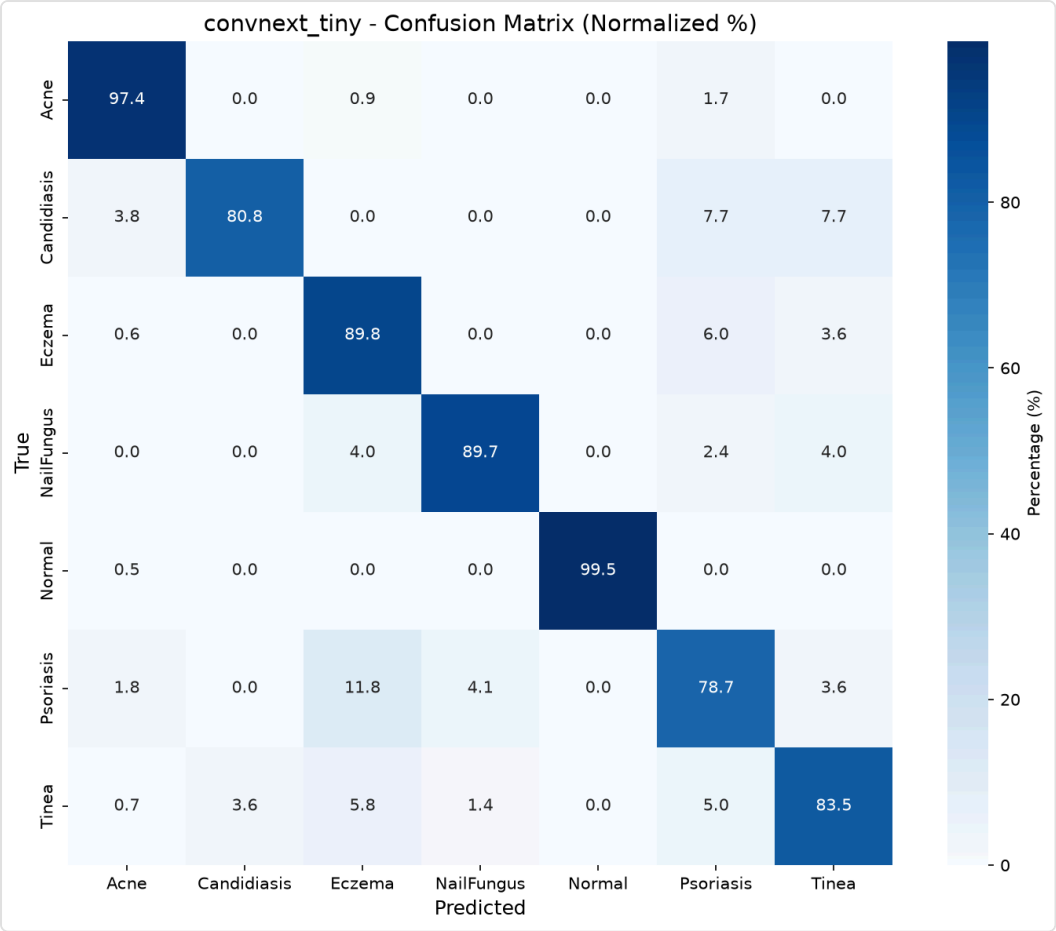

Precision-Recall Curves (Macro AP = 0.9571)
Precision
Recall
class 0 (AP = 0.9841)
class 1 (AP = 0.9841)
class 2 (AP = 0.9841)
micro-average (AP = 0.9841)
macro-average (AP = 0.9571)

9. Confusion Matrices

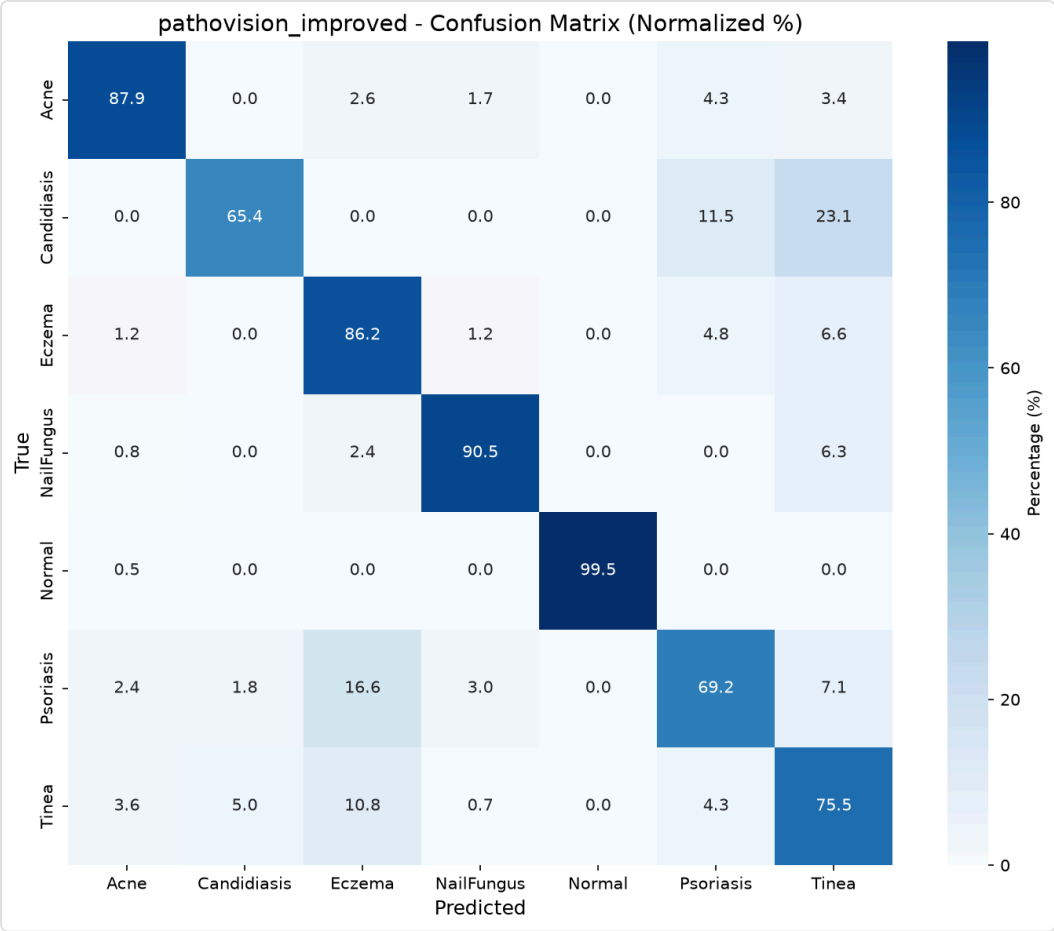
9.1 M2 — Swin Transformer



9.2 M3 — ConvNeXt Tiny



9.3 M4 — EfficientNetV2-S


pathovision_improved - Confusion Matrix (Normalized %)
True
Predicted
Acne
Candidiasis
Eczema
NailFungus
Normal
Psoriasis
Tinea
87.9 0.0 2.6 1.7 0.0 4.3 3.4
0.0 65.4 0.0 0.0 0.0 11.5 23.1
1.2 0.0 86.2 1.2 0.0 4.8 6.6
0.8 0.0 2.4 90.5 0.0 0.0 6.3
0.5 0.0 0.0 0.0 99.5 0.0 0.0
2.4 1.8 16.6 3.0 0.0 69.2 7.1
3.6 5.0 10.8 0.7 0.0 4.3 75.5
Percentage (%)
0
20
40
60
80

10. Evidence on Disk

```
D:\pathovision\evidence\
├── csvs/
│   ├── M2_swin_predictions.csv           (927 rows, 203 KB)
│   ├── M3_convnext_predictions.csv        (927 rows, 203 KB)
│   ├── M4_improved_predictions.csv        (927 rows, 203 KB)
│   ├── M4_efficientnetb2_predictions.csv  (927 rows, 209 KB)
│   └── M6_ensemble_predictions.csv        (927 rows, 203 KB)
├── figures/
│   ├── M2_swin/figures/    (ROC, PR, CM abs, CM norm)
│   ├── M3_convnext/figures/ (ROC, PR, CM abs, CM norm)
│   ├── M4_improved/figures/ (ROC, PR, CM abs, CM norm)
│   └── M4_balanced/figures/ (ROC, PR, CM abs, CM norm)
├── reports/
│   ├── summary_results.csv
│   └── mcnemar_test.txt
└── manifest.csv

Checkpoints:
├── outputs/swin/checkpoints/best.pt      (229 MB)
├── outputs/convnext/checkpoints/best.pt  (222 MB)
├── outputs/improved/checkpoints/best.pt  (101 MB)
└── outputs/checkpoints/best.pt           (52 MB)

Dataset:
└── datasets/data/
    ├── train/ (7 classes, 7,376 images)
    ├── val/   (7 classes, 921 images)
    └── test/  (7 classes, 927 images)
```

— End of Report —

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